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DEPARTMENT OF COMPUTER SCIENCE AND DESIGN

AND

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – CYBER SECURITY

Report on Industrial Visit - Vigyanlabs Innovations Pvt. Ltd., Mysuru on February 28, 2026

Vigyan labs is an innovation-driven organization focused on profitable sustainability through green technologies. Core Innovations: Inventors of Intelligent Power Management (IPM+) technology. Their Key Products is. An edge-first sovereign cloud and the world's first AI-powered Micro Data Centers. The IPM+: A unified endpoint management solution that reduces energy consumption by up to 50%. AI Services: Model Development as a Service (MDaaS), custom NLP, and Computer Vision solutions.

Objectives of the Visit:

- To provide Practical Exposure and Technical Insights for Students.
- Observed secure cloud architectures and Default VPC (Virtual Private Cloud) configurations, Asset Management and Patch Management within unified monitoring systems to mitigate security risks.
- Explored Human-Computer Interaction (HCI) through unified monitoring dashboards and AI-Help Desk interfaces. Understood the design of Modular Micro Data Centers that prioritize both efficiency and aesthetic scalability, green computing, sustainable cloud infrastructure, and AI-driven data centre management.
- The Learning Outcomes are Sustainable Computing, Gained insights into achieving a Power Usage Effectiveness (PUE) score of 1.2, significantly lower than traditional hyperscalers AI & Cloud Integration: Witnessed one-click GenAI deployment and GPU cloud compute services used for training Large Language Models (LLMs). Industry 4.0 Understood how IoT and unified monitoring drive 20% energy savings in industrial operation.

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Key Highlights

IPM+ System Architecture Overview: The core technology observed was IPM+ (Intelligent Power Management), a patented platform designed for unified endpoint management and green computing.

The architecture follows a Client-Server model augmented by Edge AI. The Intelligent Agent (Endpoint Layer) Design: A lightweight, non-intrusive service running on client devices (laptops, servers, ATMs).

Function: It uses Application Sensors to monitor user activity (keyboard/mouse) and application load without violating privacy. **Edge AI:** Unlike traditional agents that constantly the IPM+ agent hosts a local instance of the PowerMind AI engine. This allows it to make real-time power-saving decisions (e.g., throttling CPU, dimming screens) locally, even without internet connectivity.

PowerMind AI Engine (The Core) Algorithm: A Machine Learning-based engine that learns user behavior patterns (e.g., "User A takes a break between 1:00 PM and 1:30 PM").

Optimization: It constructs a hyper-personalized power policy for every device, ensuring energy is saved only when the device is truly idle or underutilized. **Server/Cloud Console (Management Layer)**

Unified Dashboard: A centralized web console (SaaS or On-Premise) that aggregates data from thousands of agents. **Communication Protocol:** Agents communicate with the server via secure HTTPS/TLS channels, sending telemetry data (heartbeats) and receiving policy updates.

The names of the Faculty Coordinators who accompanied the Students are:

Role	Faculty Name	Designation	Department
Faculty Coordinator	Prof. Razikha Amreen M I	Assistant Professor	CSE - Cyber Security
Faculty Coordinator	Prof. Shambhavi KA	Assistant Professor	CSE-Cyber Security

Batch Details:

Date of Visit	28/02/2026
Total Students	38
Departments	CSE - Cyber Security & Computer Science Design
Arrival Time	09:30 AM at Vigyanlabs, Mysuru

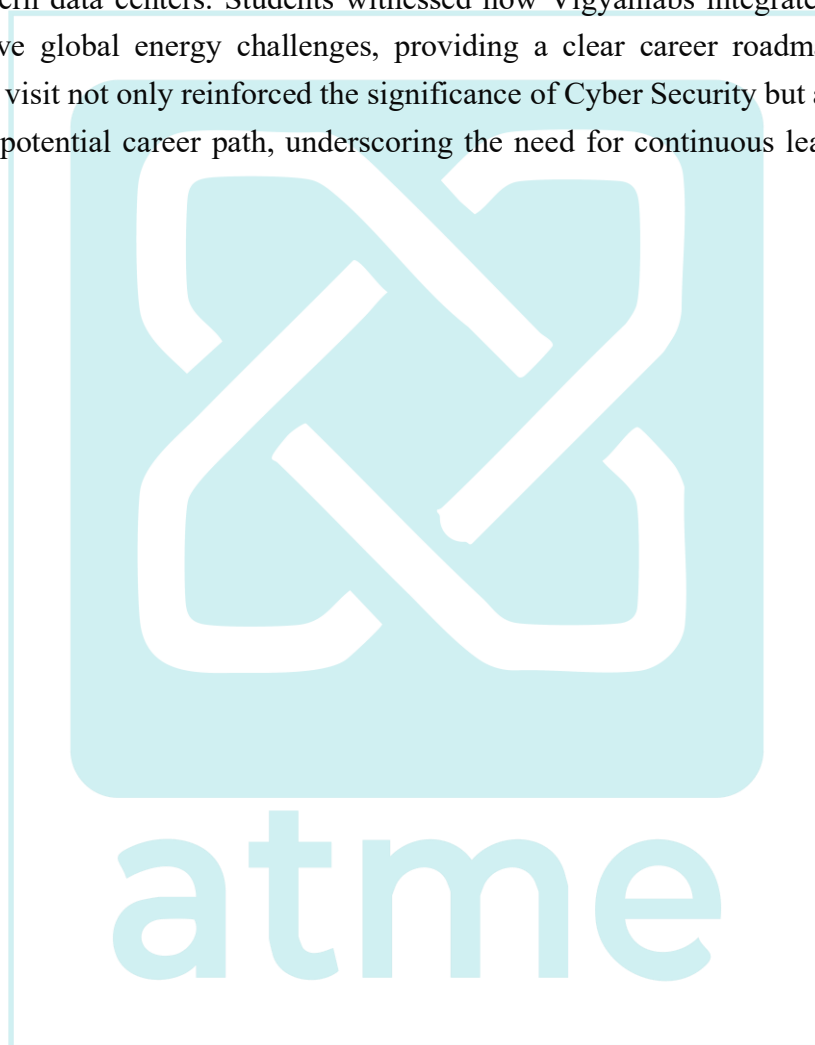
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Interactive Q&A Session

Following the tour, an interactive Q&A session was conducted. Students were encouraged to ask questions about career opportunities in Cyber Security, current industry trends, and best practices for securing personal and organizational data. Experts provided career guidance.

Conclusion

The visit bridged the gap between theoretical concepts of Green Computing and their real-world application in modern data centers. Students witnessed how Vigyanlabs integrates AI and sustainable engineering to solve global energy challenges, providing a clear career roadmap in sustainable IT infrastructure. This visit not only reinforced the significance of Cyber Security but also inspired students to consider it as a potential career path, underscoring the need for continuous learning in this rapidly evolving field.



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PHOTOS during the visit.

